SCI-CAL

TolTEC Timestream Calibration

Scientific Basis, Assumptions, and Validation

Scientific contract version 0.1

Rationale revision 0.4 – DRAFT

2026-08-20

A science-team explanation of the calibration chain, its physical meaning, uncertainty model, validity limits, and evidence still required. The separate SCI-CAL engineering contract remains the normative conformance authority.

Scientific owner: Grant Wilson

Executive summary

A TolTEC detector timestream is placed on a top-of-atmosphere, point-source-equivalent flux-density scale by applying one detector-specific calibration factor from the selected observation-specific array property table (APT), whose calibration descends from Beammap calibration evidence, and one atmospheric correction for the target sample. For detector (d), time (t), observation (o), and array (a(d)), the organizing equation is

$$d_d^{\mathrm{cal}}(t) = F_{od}^{\mathrm{APT}}\, C_{a(d)}[\tau_{225,o}(t), X_o(t)]\, d_d^{\mathrm{in}}(t). \tag{1}$$

Here d^{in} is the upstream-defined ordinary `xs` calibration input, F^{APT} is the selected row's `flxscale`, and C_a is the inverse atmospheric transmission for the target observation. The result is a point-source-equivalent, beam-peak-normalized amplitude intended to produce mJy beam^{-1} when the downstream map and filtering response has unit peak, or has been explicitly renormalized to unit peak.

The quantity τ_{225} is dimensionless zenith optical depth measured at 225 GHz, and X is the sample airmass. Their broadband, array-dependent mapping is discussed in Section 4.

The selected `flxscale` is applied exactly once. If an explicitly approved TolProj child transformation has already embodied a pointing-derived correction in the selected child APT, that ancestry is recorded but the correction is not multiplied again. None of `responsivity`, `sens`, a parent `flxscale`, or an opaque `fcf` is an additional absolute signal multiplier.

Intended science model	Equation (1), with one selected detector factor and the target sample's atmospheric correction.
Numerical atmosphere status	The materials available for this draft do not contain the operator nodes, ordinate orientation, or numeric support. Numerical atmosphere evaluation and calibrated numerical output are not supported by this draft.
Implementation status	Not assessed. This rationale neither audits nor certifies a Citlali implementation.
Observational status	Not validated here. Atmosphere fidelity, repeatability, and absolute source recovery require separate evidence.

Sample weights describe measurement uncertainty conditional on a fixed calibration state unless a documented joint covariance includes calibration, atmosphere, pointing, passband, beam, and response systematics. Common terms generally do not average down as independent samples. Formal machine states, identity mechanics, exact requirements, and pathological tests remain in the engineering contract and are traced in the appendices.

Contents

1 What is being calibrated?

SCI-CAL v0.1 is restricted to the ordinary `xs` detector stream. The selected observation-specific APT and its associated calibration lineage identify the source-calibration row used for each target detector. An ambiguous or unsupported match produces no calibrated value for that detector.

The contract begins with $d_d^{\mathrm{in}}(t)$, the signal after every upstream additive operation that defines the calibration input. SCI-CAL does not estimate a baseline. If the governing upstream convention is $d^{\mathrm{in}} = x^{xs} - b$, then b, its units, support, and uncertainty belong to that definition. This ordering matters because time-dependent multiplication need not commute with baseline estimation or filtering.

The materials available for this draft call `xs` the ordinary science stream but do not state its detector observable, physical unit, sign, normalization, preprocessing history, or linear operating domain. This draft therefore does not invent a frequency-shift interpretation, sign, or universal responsivity transfer.

> **Open Scientific Decision: Input signal and baseline**
> The relevant scientific owner must define and approve a scientific definition of ordinary `xs`: physical observable, unit, sign, prior normalization and additive processing, stream scope, and the loading, tune, and signal domain over which a multiplicative calibration is valid. The same decision must say whether an upstream baseline is not applicable, already removed, or supplied explicitly.

Polarization calibration, auxiliary measured streams, surface-brightness or temperature calibration, extended-source calibration, and integrated photometry do not inherit Equation (1); they are outside v0.1.

2 The physical calibration model and reduction chain

A minimal generative picture separates detector response from atmospheric attenuation. For a point-source-equivalent top-of-atmosphere signal $s_d(t)$, the input can be represented schematically as

$$d_d^{\mathrm{in}}(t) = R_{od} T_{a(d)}(t) s_d(t) + n_d(t), \tag{2}$$

where R is an abstract detector response in input units per $\mathrm{mJy\,beam^{-1}}$, T_a is transmission, and n is measurement noise. The central equation is the inverse operation when $F^{\mathrm{APT}} = R^{-1}$ and $C_a = T_a^{-1}$. Here R is not the APT field `responsivity`. This motivates factor direction; it does not establish the missing physical definition of `xs` or the derivation of R.

SCI-CAL lies between upstream definition of the signal, APT, opacity, and airmass and downstream filtering, weighting, mapmaking, and inference. The relevant ordering cases are

$$\begin{aligned} L_d[F_d x_d] &= F_d L_d[x_d], && \text{fixed scalar, single-detector linear operator,} \\ P[F\boldsymbol{x}] &\neq F P[\boldsymbol{x}], && \text{detector-mixing operator in general,} \\ L_d[C_d(t) x_d(t)] &\neq C_d(t) L_d[x_d(t)], && \text{sample-dependent atmosphere in general.} \end{aligned} \tag{3}$$

Common-mode removal and PCA mix detectors; a PCA basis may also be data-dependent. The materials available for this draft do not state the complete governed ordering relative to these operations, temporal filtering, or baseline estimation. They do establish that conditional uncertainty, synthetic injections, and legitimate response companions transform on the same support with the same realized multiplier. Exact pipeline ordering requires an owner-approved boundary statement before observational transfer functions can be interpreted.

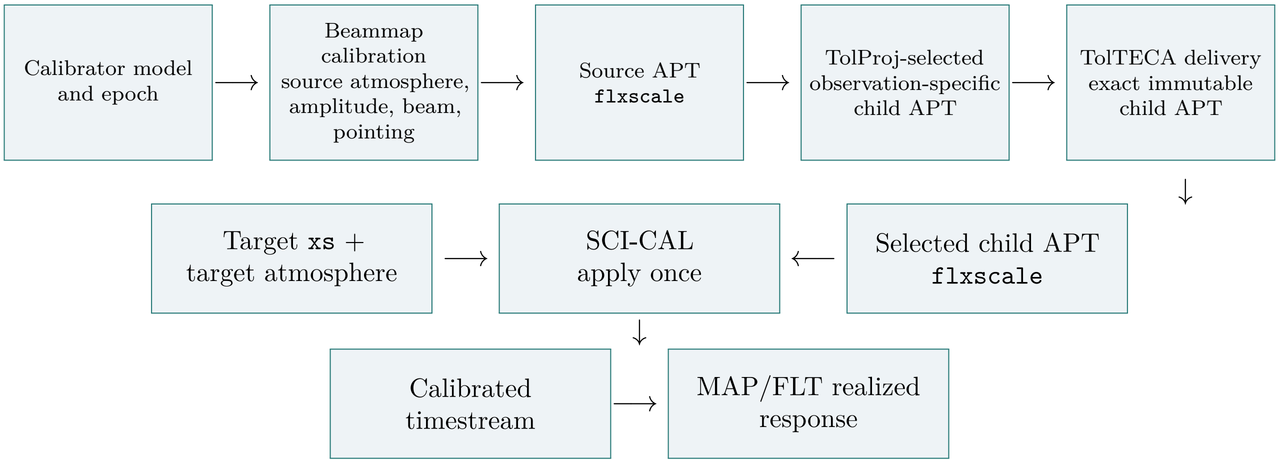

Figure 1: Scientific lineage from calibrator evidence to target-observation calibration. Beammap calibration produces the source APT factor; TolProj selects or creates the observation-specific child APT; TolTECA delivers that exact immutable artifact without changing its value or meaning; SCI-CAL consumes its selected `flxscale` and applies the target atmosphere once.

3 From a calibrator Beammap to the target observation

Beammap analysis measures detector response to a calibrator whose top-of-atmosphere flux model is adopted for the relevant epoch and photometric convention. Conceptually,

$$F_{od}^{\mathrm{APT}} \sim \frac{\text{adopted top-of-atmosphere calibrator flux density}}{\text{measured calibrator response of detector } d}. \tag{4}$$

The exact amplitude estimator, atmosphere treatment, beam/template normalization, pointing correction, spectral convention, and unit direction must be specified by the factor's generating lineage; this explanatory relation is not a substitute for that contract.

Beammap calibration and source-APT production own the calibrator model, the source-observation atmosphere, the Beammap amplitude and beam/template fit, the pointing treatment in source calibration, and the derivation and meaning of source-APT `flxscale`. TolProj owns target-to-source detector association, selection or creation of the observation-specific child APT, and only explicitly approved child transformations. TolTECA owns delivery of the exact immutable selected child artifact and its identity; delivery changes neither the child value nor its producer- and transformer-owned scientific meaning. SCI-CAL consumes that delivered child row as the value boundary, applies its `flxscale` once, and applies target atmosphere. The downstream MAP/FLT response owner owns the realized mapmaker, gridding, kernel, and filter response.

This follows the producer–transformer–delivery–consumer rule: the producer owns a quantity's meaning, a transformer owns only its approved transformation and lineage, delivery preserves the exact selected artifact, and a consumer applies the selected value without reinterpretation. If $F^{\mathrm{child}} = PF^{\mathrm{parent}}$, the right side records ancestry; neither P nor F^{parent} is then applied to the target signal.

Every target detector needs one unique, traceable, observation-valid match to one measured APT row. Exact mechanics remain in the engineering contract; an unmatched or ambiguous detector is excluded rather than calibrated from row position or a plausible default.

Table 1: Roles of calibration-related quantities.

Quantity	Physical role	Source	Signal factor?	Correlation or uncertainty scope
`flxscale`	Selected absolute detector scale	Selected observation-specific child APT; derived from source-APT/Beammap calibration	Yes, once	Detector plus shared calibrator ancestry
Pointing correction	Correction in APT ancestry	Beammap source calibration or an explicitly approved child transformation	No, if embodied	Detector/observation correlated
`responsivity`	Relative response or diagnostic	Selected APT lineage	No	Transfer role unresolved
`sens`	Sensitivity or approximate weight	Beammap analysis	No	Detector/cohort; not total error
Target C_a	Inverse transmission for target sample	Opacity, airmass, array operator	Yes, once	Time, observation, and array correlated
Compatibility `fcf`	Declared decomposition of known roles	Compatibility boundary	Not independently	Inherits constituent scopes

Open Scientific Decision: Calibration derivation and transfer
The materials available for this draft do not state the exact construction of `flxscale`, calibrator model and epoch, embodied source-atmosphere/beam/pointing/spectral operations, or how tune state, loading, focus, time, and detector responsivity transfer from calibrator to target. The relevant scientific owner must define and approve the generating contract and decide whether transfer is embodied in the selected child `flxscale`, made unnecessary by `xs`, negligible over a declared domain, or retained as a correlated systematic.

4 Atmospheric extinction correction

The atmosphere attenuates the astronomical signal. Zenith optical depth at 225 GHz, τ_{225}, and sample airmass $X(t)$ locate each sample within an adopted per-array atmosphere/passband model:

$$\ell_o(t) = \tau_{225,o}(t)X_o(t), \qquad H_a(t) = \mathcal{O}_a[\ell_o(t)], \qquad (T_a, C_a) = \begin{cases} (H_a, H_a^{-1}), & H_a \text{ is transmission,} \\ (H_a^{-1}, H_a), & H_a \text{ is correction.} \end{cases} \tag{5}$$

The quantity τ_{225} is not itself the TolTEC-band opacity. The operator performs the declared broadband mapping. If it instead tabulates correction, that declared ordinate is interpolated first and transmission is its reciprocal; interpolating transmission and correction are different approximations.

The opacity source owns temporal validity and any interpolation between valid bracketing measurements. ALIGN/AST owns sample airmass and its domain. SCI-CAL uses the intersection with operator support; it does not extrapolate or invent an elevation limit. Detailed equations are in Appendix A.

Open Scientific Decision: Authoritative atmosphere operator
The materials available for this draft do not contain the per-array nodes and ordinates, ordinate orientation, units, numeric support, seam convention, passband identity, spectral and photon-versus-energy weighting, generating model, or content identifier. It also does not establish whether that record is absent project-wide or merely absent from the materials reviewed. The relevant scientific owner must classify the status and define and approve one complete immutable record. Until then correction curves, numerical examples, an approximation-error plot, numerical atmosphere evaluation, and calibrated numerical output are not supported. Supplying the record enables structural evaluation; it does not validate atmospheric fidelity.

The calibration path defined by the present contract applies only for $0 \le \tau_{225} \le 0.15$. A coherent segment entering $0.15 < \tau_{225} \le 0.25$ is outside the v0.1 science-calibration range and is retained only for explicitly uncalibrated engineering use. Larger, malformed, or unsupported values are outside scope. The thresholds are adopted v0.1 policy limits, not laws of atmospheric physics and not the numerical support of the missing operator. The materials reviewed do not document their scientific rationale, the authority that defines coherent segments, or the split policy; those matters remain open under Q07.

5 Photometric convention and downstream response

The intended output is a top-of-atmosphere, point-source-equivalent, beam-peak-normalized amplitude. "Beam" denotes the originating Beammap response normalization; it is neither pixel area nor an approved definition of surface brightness or integrated flux.

For a source whose flux density is S mJy, originating unit-peak template P_B, and realized map-maker/gridding/filter response (L),

$$m_{\text{peak}} = \rho_L S \text{ mJy beam}^{-1}, \qquad \rho_L = [LP_B](\mathbf{0}). \tag{6}$$

The literal peak equals S only for $\rho_L = 1$, or after documented renormalization. A valid smoothing kernel can broaden a point source and reduce its peak; retaining the old label without response correction would then misstate flux.

TolTEC is broadband. Monochromatic mJy requires a reference frequency or wavelength, reference spectrum, calibrator-spectrum treatment, passband weighting, and color-correction convention. The approved modeled passband set does not establish normalization, detector/network variation, uncertainty, generation recipe, or photon-versus-energy weighting. The same spectral convention must connect `flxscale` and atmosphere consistently.

Open Scientific Decision: Broadband photometric convention
The relevant scientific owner must define and approve the per-array reference frequency or wavelength, reference spectrum, calibrator treatment, color-correction scope, passband weighting, and whether the atmosphere operator is array-average, network-specific, or detector-specific. Until then the intended point-source convention is identified, but its complete monochromatic mJy meaning is unresolved.

6 Uncertainty: measurement noise is not total calibration error

For fixed calibration, atmosphere, detector association, and response, let $A = \text{diag}(M_j)$, with $M_j = F_{od}^{\text{APT}} C_{a(d)}(t)$. A supplied conditional covariance obeys

$$C_{\text{cal}|\theta} = AC_{\text{in}|\theta}A^{\mathsf{T}}, \qquad \sigma^2_{\text{cal},j} = M_j^2 \sigma^2_{\text{in},j}, \quad w_{\text{cal},j} = w_{\text{in},j}/M_j^2. \tag{7}$$

This preserves supplied correlations. Missing uncertainty is unavailable, never zero. Calibration-state uncertainty is different. A fractional scale uncertainty (s) common to affected samples contributes

$$C_{\text{common}} = s^2 \boldsymbol{\mu}\boldsymbol{\mu}^{\mathsf{T}}, \tag{8}$$

which does not decrease merely because more identically affected samples are averaged.

Table 2: Science-facing uncertainty budget. “Not established” means the materials available for this draft do not state a present numerical product.

Quantity	Typical correlation scope	In sample weight?	Current status	Interpretation
Measurement noise	Sample/detector, possibly correlated	Yes, if supplied	Conditional input allowed	Propagated by ACA^T
Detector `flxscale` fit	Detector/Beammap cohort	No by default	Not established	Calibration systematic
Calibrator scale/model	Observations and/or band	No	Not established	Does not average down
Pointing correction	Detector/observation	No	Not established	Embodied once
Atmosphere, WVR, operator	Time, observation, array	No	Operator unresolved	Model systematic
Passband/source spectrum	Array/network/detector	No	Convention unresolved	Broadband systematic
Beam/filter response	Product/reduction state	No	Downstream-owned	Needed for literal peak
`sens` approximation	Detector/cohort	Only if declared	Not established	Never total error

The sources define compliant meaning but do not establish which numerical components a present Citlali product supplies. That is later implementation evidence, not a fact inferred here.

7 Validity limits and what a science user sees

Table 3: Meaning of calibration status for science users.

Status	Calibrated value?	Conditional weight?	Interpretation and action
Structurally complete within nominal domain	Yes, once numeric inputs are supported by approved definitions	If supplied	Algebra and inputs complete; atmosphere fidelity and on-sky performance remain separate.
Calibration disabled	No	No calibrated weight	Intentional uncalibrated mode; do not relabel input.
Unsupported stream or unit	No	No	A separate scientific contract is required.
Invalid/missing atmosphere	No	No	Correct the owning input or exclude segment.

Status	Calibrated value?	Conditional weight?	Interpretation and action
Outside operator support	No	No	No extrapolation, clamp, or legacy fallback.
Engineering-only opacity	No v0.1 value	No	Retained only for explicitly uncalibrated engineering use.
Conditional uncertainty unavailable	Signal may exist	No	Do not interpret missing uncertainty as zero.
Total uncertainty incomplete	Signal may exist	Conditional only	Report noise separately from systematics.
Response basis incomplete	Signal may exist	Conditional only	Withhold literal peak meaning.

The engineering state `science-qualification-eligible` is rendered here as "structurally complete but not yet scientifically validated."

8 What has to be validated, and what each validation establishes

1. **Structural calibration correctness.** Factor direction and once-only challenges; unit and support checks; unique detector to APT association; zero opacity, node, seam, monotonicity, and no extrapolation behavior; conditional covariance scaling; and response bookkeeping. Software-oriented permutation, duplicate-key, replay, sentinel, and application-count tests remain in the engineering contract.
2. **Atmosphere-model representation fidelity.** Compare the finite operator with its named higher-fidelity generating calculation over each array's declared opacity, airmass, spectral, and passband domain. The prior approximately one-percent figure is a proposed target, not an achieved result.
3. **Observational calibration performance.** Recover repeated calibrators and independent sources by array; separate repeatability from absolute recovery; test residuals versus opacity, airmass, elevation, time, Beammap age, tune state, loading, detector, and network; inject sources through the realized response; and retain correlated calibrator covariance. Prior approximately five-percent repeatability and five-to-ten-percent absolute figures are proposed targets only.

Any claim must identify source population, atmosphere domain, array coverage, response configuration, covariance model, sample size, and criterion. Passing one layer never supplies evidence for another.

9 Scientific owner decisions required

Table 4: Decision register for science-rationale revision r0.4.

ID	Issue	Evidence or decision needed	Consequence while open
Q01	Physical `xs` definition	Observable, unit, sign, preprocessing, normalization, scope, and linear domain	Input/factor units incomplete

ID	Issue	Evidence or decision needed	Consequence while open
Q02	Baseline and ordering	Affine convention and order relative to common-mode/PCA/filtering	Noncommuting operations ambiguous
Q03	`flxscale` derivation	Calibrator model/epoch, estimator, atmosphere, pointing, beam, spectrum, direction, uncertainty	Factor ancestry incomplete
Q04	Calibrator-target transfer	Time, tune, loading, focus, detector validity, and role of child factor or `xs`	Transferability unestablished
Q05	Broadband convention	Reference frequency/spectrum, calibrator and color treatment, passband weighting/variation	Monochromatic mJy incomplete
Q06	Atmosphere operator	Nodes, ordinates, orientation, support, weighting, model, content identity; classify project-wide versus reviewed-material absence	Blocks contract-supported numerical calibration
Q07	Opacity/segment policy	Rationale for 0.15/0.25, segment owner, split policy, evidence for change	Policy unexplained
Q08	Numerical uncertainty	Which conditional/systematic terms are quantified and propagated	Total error unavailable
Q09	Criteria for validated science use	Evidence thresholds and population/support for fidelity, repeatability, absolute recovery	No achieved science claim

References

[1] *SCI-CAL Detector Calibration, Atmospheric Extinction, and Signal Transfer Scope Brief*, owner-approved 2026-08-16.

[2] *SCI-CAL Independent Scientific Core*, with v0.1 supersession cover.

[3] *SCI-CAL Sanitized Conventions and Ownership*, owner-approved 2026-08-16.

[4] *TolTEC v1 Modeled Array Passband Authority Manifest*, bounded reference with unresolved semantics.

The sanitized conventions/ownership reference records the lineage of its approved extract. No broader source in that lineage was independently opened or used as an author input. No calibrator-model, atmosphere-model, `xs`-definition, or broadband photometric-convention source was supplied as approved authority. Those are decisions Q01 and Q03–Q06 rather than unnamed citations.

A Notation and formal derivations

Table 5: Notation retained for formal traceability.

Symbol	Meaning
(o,d,t,a(d))	Observation, detector, sample time, detector array
$d^{\mathrm{in}}, d^{\mathrm{cal}}$	Upstream-defined input and calibrated output
F_{od}^{APT}	Selected row's oriented absolute `flxscale`
τ_{225}, X, ℓ	Zenith opacity, airmass, line-of-sight coordinate
H_a, T_a, C_a	Interpolated ordinate, transmission, correction
M_j, A	Realized scalar multiplier and diagonal operator
$C_{\mathrm{in}}, C_{\mathrm{cal}}, C_\theta$	Conditional input/output and nuisance covariance
P_B, L, ρ_L	Beammap template, downstream response, peak response

If upstream signal definition includes a baseline,

$$d_j^{\mathrm{in}} = x_j^{xs} - b_j. \tag{9}$$

SCI-CAL does not estimate b_j. On a declared atmosphere interval,

$$u = \frac{\ell - \ell_{a,k}}{\ell_{a,k+1} - \ell_{a,k}}, \tag{10}$$

$$H_a(\ell) = (1-u)H_{a,k} + uH_{a,k+1}, \tag{11}$$

$$(T_a, C_a) = \begin{cases} (H_a, H_a^{-1}), & H_a \text{ is transmission,} \\ (H_a^{-1}, H_a), & H_a \text{ is correction.} \end{cases} \tag{12}$$

The declared ordinate is interpolated before reciprocal. Exact nodes are recovered, seams continuous, $T_a(0) = C_a(0) = 1$, transmission decreases, correction increases, and extrapolation is unauthorized.

For fixed state θ,

$$\mathbb{E}[\boldsymbol{d}^{\mathrm{cal}} \mid \theta] = A\mathbb{E}[\boldsymbol{d}^{\mathrm{in}} \mid \theta], \qquad C_{\mathrm{cal}|\theta} = AC_{\mathrm{in}|\theta}A^{\mathsf{T}}. \tag{13}$$

For small nuisance uncertainty with $H_{jr} = \partial \ln |M_j| / \partial \theta_r$,

$$C_{\mathrm{nuis}} \simeq \mathrm{diag}(\boldsymbol{\mu}) H C_\theta H^{\mathsf{T}} \,\mathrm{diag}(\boldsymbol{\mu}). \tag{14}$$

Shared-data calibration or baseline estimation also requires material cross covariances. Large, asymmetric, seam-spanning, or discrete uncertainty requires nonlinear samples, a joint posterior, or bounded sensitivity study.

B Exact machine states and formal conformance routing

The engineering authority distinguishes requested, effective, observation-resolved, and realized states and valid signal. Its exact named states include:

- `disabled/uncalibrated`;
- `outside-supported-calibration`;

- `engineering-only-unavailable`; and
- `science-qualification-eligible`.

It separately records conditional-uncertainty availability, total-uncertainty completeness, and response completeness. Table 3 translates these for users; it does not replace them. Appendix D maps every formal item to this rationale and its continuing engineering home.

C Provenance and document authority

The approved input content digests are:

- Scope Brief: `a3d6332dcc98bfbb638a45aea9be830edca693a4fe7e65d831b5880603c16578`;
- independent core: `106755520b048f601bc60fd04e7b6020e6fa470480ac3105fa7ba269c730a4fe`;
- passband manifest: `2756908181cc466550399ec0a869e6671de7912bd3a935f9aeebf63e3e826617`;
- conventions/ownership: `7e9a630fd183ca04bc3d8bbd21b5e801776b9aad7bd084d48fa7f2c572766520`.

The engineering contract remains normative for requirements 001–050, assumptions 001–011, and edge predictions 001–030. Rationale revision r0.4 changes genre and presentation, not the governed v0.1 algebra.

D Crosswalk to the engineering contract

The engineering contract remains normative. This compact map locates every formal assumption, requirement, and edge prediction without reproducing its full audit-oriented text in the science narrative.

Table 6: Assumption routing.

Formal IDs	Scientist-facing location	Continuing engineering home
`SCI-CAL-ASM-001`, `SCI-CAL-ASM-002`	Sections 1–2; Q01–Q02	Approved input support, ordinary channel, and binding
`SCI-CAL-ASM-003`	Section 3; Figure 1; Q03–Q04	Source APT, TolProj-selected child APT, immutable TolTECA delivery, and association
`SCI-CAL-ASM-004`	Section 1; Appendix A	Explicit affine-input convention
`SCI-CAL-ASM-005`	Sections 2–3; Q03–Q05	Complete factor lineage, direction, plane, and once-only use
`SCI-CAL-ASM-006`	Section 4; Q06–Q07	Atmosphere inputs, time support, and policy
`SCI-CAL-ASM-007`	Section 6	Conditional-state covariance
`SCI-CAL-ASM-008`	Section 5	Originating and downstream response ownership
`SCI-CAL-ASM-009`	Sections 4, 7; Q07	Coherent segment quality
`SCI-CAL-ASM-010`	Sections 4–5; Q05	Passband limitations and photometric compatibility
`SCI-CAL-ASM-011`	Sections 4, 9; Q06	Unresolved numeric operator authority

Table 7: Requirement routing.

Formal IDs	Scientist-facing location	Continuing engineering home
SCI-CAL-REQ-001, SCI-CAL-REQ-002, SCI-CAL-REQ-003, SCI-CAL-REQ-004, SCI-CAL-REQ-005	Sections 1, 7; Q01–Q09	Scope, typed state, authority snapshot, validity, and one-way resolution
SCI-CAL-REQ-006, SCI-CAL-REQ-007, SCI-CAL-REQ-008, SCI-CAL-REQ-009, SCI-CAL-REQ-010, SCI-CAL-REQ-011, SCI-CAL-REQ-012	Section 3; Figure 1	Acquisition/source/child identity, binding, association, delivery, and lifetime
SCI-CAL-REQ-013, SCI-CAL-REQ-014, SCI-CAL-REQ-015, SCI-CAL-REQ-016	Sections 2–3; Q03–Q04	Generating record, child transform, transfer domain, and once-only multiplier
SCI-CAL-REQ-017, SCI-CAL-REQ-018, SCI-CAL-REQ-019, SCI-CAL-REQ-020	Section 3; Table 1	Relative roles, compatibility, producer/transformer/delivery/consumer lineage
SCI-CAL-REQ-021, SCI-CAL-REQ-022, SCI-CAL-REQ-023, SCI-CAL-REQ-024, SCI-CAL-REQ-025, SCI-CAL-REQ-026	Section 4; Appendix A	Atmosphere meaning, temporal/operator support, invariants, and Q06-only closure
SCI-CAL-REQ-027, SCI-CAL-REQ-028, SCI-CAL-REQ-029, SCI-CAL-REQ-030, SCI-CAL-REQ-031	Sections 4, 7; Q05–Q07	Opacity policy, coherent quality, passband limits, and spectral compatibility
SCI-CAL-REQ-032, SCI-CAL-REQ-033, SCI-CAL-REQ-034	Section 6	Conditional covariance, variance, weight, unavailable state
SCI-CAL-REQ-035, SCI-CAL-REQ-036, SCI-CAL-REQ-037, SCI-CAL-REQ-038, SCI-CAL-REQ-039	Sections 2, 6; Appendix A; Q02	Nuisance scopes, total covariance, companions, and local-versus-end-to-end response
SCI-CAL-REQ-040, SCI-CAL-REQ-041, SCI-CAL-REQ-042, SCI-CAL-REQ-043, SCI-CAL-REQ-044	Sections 3, 5	Producer-owned beam/template basis, realized response, and unit exclusions
SCI-CAL-REQ-045, SCI-CAL-REQ-046, SCI-CAL-REQ-047, SCI-CAL-REQ-048	Sections 3, 7; Appendix B; Q01–Q09	Claim-specific states, reasons, complete ownership/delivery lineage, and product links

Formal IDs	Scientist-facing location	Continuing engineering home
SCI-CAL-REQ-049, SCI-CAL-REQ-050	Sections 5, 8; Q01–Q09	Separate physical, transfer, broadband, response, structural, fidelity, and performance claims

Table 8: Edge-prediction routing.

Formal IDs	Scientist-facing location	Continuing engineering home
SCI-CAL-EDGE-001, SCI-CAL-EDGE-002, SCI-CAL-EDGE-003, SCI-CAL-EDGE-004, SCI-CAL-EDGE-005	Section 4; Appendix A	Zero, monotonicity, zenith, exact-node, and seam checks
SCI-CAL-EDGE-006, SCI-CAL-EDGE-007, SCI-CAL-EDGE-008, SCI-CAL-EDGE-009, SCI-CAL-EDGE-010	Sections 4, 7	Domain, temporal bracketing, engineering/outside states
SCI-CAL-EDGE-011, SCI-CAL-EDGE-012, SCI-CAL-EDGE-013	Section 3	Unity, omitted/duplicate/inverted, corrected-APT challenges
SCI-CAL-EDGE-014, SCI-CAL-EDGE-015, SCI-CAL-EDGE-016, SCI-CAL-EDGE-017, SCI-CAL-EDGE-018, SCI-CAL-EDGE-019	Section 3	Row/order/network, missing/duplicate, association/design tests
SCI-CAL-EDGE-020, SCI-CAL-EDGE-021, SCI-CAL-EDGE-022, SCI-CAL-EDGE-023, SCI-CAL-EDGE-024	Section 6	Scalar, dense, unavailable, common, and nonlinear uncertainty tests
SCI-CAL-EDGE-025, SCI-CAL-EDGE-026, SCI-CAL-EDGE-027	Section 5	Unit-peak source, response-changing kernel, beam approximation
SCI-CAL-EDGE-028	Sections 1, 7	Unsupported stream or target
SCI-CAL-EDGE-029	Section 1; Appendix A	Zero signal and affine convention
SCI-CAL-EDGE-030	Sections 3, 7	Observation-order state replay